

TEACHTRACE AI

AI-Powered Classroom Intelligence & Learning Gap Detection System

"Don't just trace the marks. Trace the learning gap."

Pak Angels GenAI & Agentic AI Cohort 11 — Mid-Program Hackathon

Team Lead: Naveen Khan | naveenkhan0111@gmail.com

The Problem: Marks Show WHO Failed, Not WHY They Struggled

✗ Reactive Grading Trap

Traditional assessment tools focus entirely on score calculation (e.g. 45/100). They treat learning errors as final grades rather than diagnostic opportunities.

⌚ Manual Analysis Bottleneck

Teachers must manually examine dozens of handwritten test papers to spot recurring mistakes — an impossible task for large classrooms.

⚙️ Unseen Hidden Causes

Scores obscure whether an error is due to conceptual gaps, repeated misconceptions, language/grammar issues, or ambiguous test question wording.

The Solution: Transforming Evidence into Actionable Teaching

1

RAG & Linguistic Analysis

Ingests student answers & reference keys to evaluate grammar, vocabulary, reasoning logic, and misconceptions.

2

Learning Gap Map

Aggregates individual errors into class-wide trend maps displaying affected students and severity tiers.

3

Targeted Reteaching

Generates 15-minute classroom intervention activities and diagnostic checks directly addressing the top gap.

4

Measured Improvement

Generates follow-up assessments and compares pre- vs post-intervention scores to prove learning growth.

Technology Stack & Architecture (Cohort 11)



Streamlit Frontend

Modular teacher dashboard with multi-page navigation, reactive state management, and gap map visualizers.



Groq & Gemini APIs

High-speed LLM inference powered by Groq Llama 3.3 (70B) for instant linguistic and misconception analysis.



LangChain & RAG

RAG retrieval framework storing reference material and textbook keys in ChromaDB for grounded diagnostics.



Streamlit Cloud

Fully configured repository (requirements.txt, .streamlit/secrets.toml) deployed for instant cloud access.

System Flow: Closed-Loop Diagnostic & Intervention Workflow

TEACHTRACE AI SYSTEM FLOW

- 1 Teacher Upload** — Student written answers (CSV/PDF) + Reference textbook notes
- 2 RAG & Linguistic Parser** — Evaluates Grammar, Terminology, Logic & Misconceptions via Groq
- 3 Learning Gap Map** — Displays class-wide gaps, affected student counts & Question Ambiguity alerts
- 4 Targeted Intervention** — Generates 15-minute reteaching plan & student sorting/explanation activity
- 5 Reassessment Engine** — Evaluates 3-question follow-up test and measures score improvement (e.g. 43% → 78%)

Teacher Dashboard

Live class overview: enrollment, urgent learning gaps by severity, and per-topic performance at a glance.



Upload & Analyze Answers

Teachers upload student responses alongside the reference answer key; RAG + linguistic analysis flags misconceptions per student.



TeachTrace AI

Classroom Intelligence & Gap Detection

Navigation

- ☐ My Classes & Dashboard
- ☒ Upload & Analyze Answers
- ☐ Learning Gap Map
- ☐ Reteaching & Interventions
- ☐ Reassessment & Improvement

Active Class

Grade 8-A — Mathematics

Core Mission: Don't just trace the marks. Trace the learning gap.

Deploy

Upload Student Written Answers & Reference Material

Upload student test responses (CSV/PDF/TXT) alongside textbook reference notes for RAG + Linguistic Analysis.

1. Student Responses

Upload Student Answer File (CSV/PDF/TXT)

Upload

200MB per file • CSV, TXT, PDF

☐ Use Demo Classroom Data (30 Students — Fractions Test)

2. Reference & Answer Key

Reference Answer Key / Learning Objectives Notes

Topic: Fraction Addition & Denominators

Core Concept: When adding fractions, students must find a common denominator before adding numerators.

Common Misconception: Adding numerators directly and denominators directly (e.g., $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$).

Correct Method: $\frac{1}{2} + \frac{1}{3} = \frac{2}{6} + \frac{2}{6} = \frac{4}{6}$

Run RAG & Linguistic Analysis

Sample Individual Student Diagnostics

student	question	answer	score	issue
---------	----------	--------	-------	-------

2 / 5

Learning Gap Map

Individual errors are aggregated into a class-wide map of root causes, affected-student counts, and severity.



TeachTrace AI

Classroom Intelligence & Gap Detection

Navigation

- ☐ My Classes & Dashboard
- ☐ Upload & Analyze Answers
- ☒ Learning Gap Map
- ☐ Reteaching & Interventions
- ☐ Reassessment & Improvement

Active Class

Grade 8-A — Mathematics

 **Core Mission:** Don't just trace the marks. Trace the learning gap.

Deploy



Classroom Learning Gap Map

Aggregated linguistic and conceptual analysis identifying *why* students are getting questions wrong.

Class-Wide Learning Gap Map

Aggregated analysis of 30 student responses indicating root causes of errors:

>  Adding Denominators Directly — Affected: 18 / 30 (60%) | Severity: High

>  Finding Least Common Multiple (LCM) — Affected: 12 / 30 (40%) | Severity: Medium

>  Fraction Terminology (Numerator vs Denominator) — Affected: 9 / 30 (30%) | Severity: Medium

>  Incomplete Explanation Steps — Affected: 7 / 30 (23%) | Severity: Low



Question Ambiguity Detector

Potential Question Ambiguity Detected!

3 / 5

Reteaching & Interventions

A 15-minute intervention plan — visual model, activity, and a misconception diagnostic — targets the top gap directly.



TeachTrace AI

Classroom Intelligence & Gap Detection

Navigation

- ☐ My Classes & Dashboard
- ☐ Upload & Analyze Answers
- ☐ Learning Gap Map
- ☒ Reteaching & Interventions
- ☐ Reassessment & Improvement

Active Class

Grade 8-A — Mathematics

 **Core Mission:** Don't just trace the marks. Trace the learning gap.

Strategy:

- Visual Area Model (5 mins):** Use a pizza or pie chart to show why $\frac{1}{2} + \frac{1}{3}$ cannot equal $\frac{2}{5}$ (show that $\frac{2}{5}$ is smaller than $\frac{1}{2}$!).
- Denominators as Unit Names (5 mins):** Explain that denominators are like labels (e.g., 1 Apple + 1 Orange cannot equal 2 Apploranges; you need a common unit 'Fruits').
- Targeted Sort Activity (5 mins):** Have students classify fraction addition problems as 'Same Denominators (Add Top)' vs 'Different Denominators (Convert First)'.



Classroom Activity: Sort & Explain

Classroom Activity: "Same Unit or Convert First?"

Instructions for Students: Classify the following fraction pairs. Do you need to convert to a common denominator first? Explain WHY.

- $\frac{1}{5} + \frac{3}{5} \rightarrow$ ☐ Add directly ☐ Convert first
- $\frac{1}{4} + \frac{1}{2} \rightarrow$ ☐ Add directly ☐ Convert first
- $\frac{2}{7} + \frac{4}{7} \rightarrow$ ☐ Add directly ☐ Convert first

Discussion Prompt: Why is $\frac{1}{4} + \frac{1}{2}$ NOT equal to $\frac{2}{6}$? Draw a diagram to justify your answer.



Misconception Diagnostic Question

Quick Check Question: Sara says: "To add $\frac{2}{5}$ and $\frac{1}{5}$, I get $\frac{3}{5}$. So to add $\frac{2}{5}$ and $\frac{1}{10}$, I should get $\frac{3}{15}$."

Is Sara correct or incorrect? Explain the specific rule Sara forgot about denominators.

Deploy

4 / 5

Reassessment & Improvement

A short AI-generated follow-up test measures whether the targeted reteaching actually closed the gap.



TeachTrace AI

Classroom Intelligence & Gap Detection

Navigation

- ☐ My Classes & Dashboard
- ☐ Upload & Analyze Answers
- ☐ Learning Gap Map
- ☐ Reteaching & Interventions
- ☒ Reassessment & Improvement

Active Class

Grade 8-A — Mathematics

 **Core Mission:** Don't just trace the marks. Trace the learning gap.

Deploy



Reassessment & Improvement Measurement

Measure whether student understanding improved following targeted intervention.

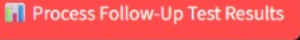
1. AI-Generated Follow-Up Assessment (3 Questions)

Short evaluation focused exclusively on the detected learning gap:

Q1. Calculate $\frac{2}{5} + \frac{1}{4}$. Show every step of your common denominator conversion.

Q2. Explain why adding $\frac{1}{3} + \frac{1}{3}$ results in $\frac{2}{3}$, but $\frac{1}{3} + \frac{1}{2}$ does NOT result in $\frac{2}{5}$.

Q3. Identify the mistake in Ahmed's answer: " $\frac{3}{8} + \frac{2}{8} = \frac{5}{16}$ ".



5 / 5

Project Team & Acknowledgements (Pak Angels Cohort 11)

Naveen Khan

Team Lead & AI Architect

naveenkhan0111@gmail.com

Saima Noor

Core Developer

saimaattiq27@gmail.com

Muhammad Aqib Aziz

Core Developer

maqib8373@gmail.com

Maheen Arshad

Core Developer

maheenarshad2607@gmail.com

Dua Maheen

UI/UX Specialist

maheenkdua@gmail.com

Ahmad Arif

QA & Evaluation Lead

f2025se037@niit.edu.pk